

SOFTWARE ENGINEERING-Lab 03

Name: Dhanush S Shekhar

SRN: PES1UG24CS662

Architecture Selection — Content Creator Milestone & Escrow Platform

Architecture Selection

We chose **Layered Architecture** for the Content Creator Milestone & Escrow Platform.

Reason 1 — Separation of Responsibilities

The system can be separated into Presentation, Business and Data layers. The Presentation Layer handles interactions with content creators and client sponsors, while the Business Layer manages contracts, milestones, deliverable submission, watermarking and payment processing. The Data Layer stores contract, milestone, payment and media information. This separation makes the system easier to understand and maintain.

Reason 2 — Clear Management of the Escrow Workflow

Layered architecture allows the milestone and payment rules to remain within the Business Layer. The system can ensure that a milestone is approved by the client sponsor before the Payment Service requests payment processing. This supports the core escrow workflow in which payment is released only after sponsor approval.

Security Advantage

The architecture allows authorization and access-control checks to be handled in the Business Layer before users access contracts, drafts, milestones or payment information. The Presentation Layer does not directly access the database, helping prevent unauthorized direct access to stored information. This supports the security requirement that only authenticated and authorized users associated with a contract can access its information.

Performance Benefit

The Watermark Processing Component can be separated from the user-interface component so that media processing is handled by a dedicated component. This allows watermarking operations to be optimized independently while keeping the user interface responsible only for interaction. This is relevant because the system requires at least 95 out of 100 watermarking operations to complete within five seconds.